STREAMLIT TRADING AND ANALYTICS PLATFORM

AN END-TO-END MODULAR ALGORITHMIC DESK AND OPTIONS FRAMEWORK



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ABOUT

This document provides a reference guide to the metrics, performance measures, risk measures, and terminology used throughout the Project Quant Backtest Engine.

The purpose is to make the backtest results easier to interpret and understand rather than simply presenting a final profit or loss. Each metric helps evaluate a different aspect of a trading strategy — including returns, profitability, consistency, drawdowns, risk, trade quality, and overall strategy performance.

The document explains what each metric means, how it is calculated or interpreted within the backtest engine, and what it tells us about the underlying strategy.
It covers the terminology used in both:

- Stock Backtesting
- Stock + Options Backtesting

Metrics are grouped into logical categories so that the results can be evaluated from multiple perspectives rather than relying on a single performance number.

Important : A strategy should not be considered successful based solely on its total return. A proper evaluation should also consider factors such as drawdown, risk-adjusted returns, win/loss characteristics, consistency, capital utilization, and trade-level performance.

BACKTEST ENGINE



Net Profit/Loss

Definition

Net Profit/Loss represents the total monetary gain or loss generated by a trading strategy after all completed trades during the selected backtesting period. It is the most fundamental performance metric and indicates whether the strategy was profitable overall.

A positive value indicates that the total profits exceed the total losses, while a negative value indicates the opposite.

Formula

Net Profit/Loss = Final Capital-Initial Capital

For an individual trade:
Long Position : (Exit Price-Entry Price)×Quantity
Short Position : (Entry Price-Exit Price)×Quantity

Purpose

Net Profit/Loss measures the overall financial outcome of a trading strategy and serves as the primary indicator of profitability. It answers a simple question:

Did the strategy make or lose money?
Although it is often the first metric traders evaluate, it should always be analysed alongside risk-adjusted performance metrics.

Limitations

Net Profit/Loss alone does not provide information about:

- Risk undertaken to generate the returns.
- Consistency of profits.
- Capital efficiency.
- Drawdowns experienced during the strategy.

Two strategies may generate identical Net Profit but have vastly different risk profiles. Therefore, this metric should always be evaluated together with Maximum Drawdown, Profit Factor, Expectancy, and Sharpe Ratio.

Stock vs Stock + Options

The interpretation of Net Profit/Loss remains the same for both Stock and Stock + Options Backtesting.

- **Stock Backtesting** : Profit is calculated from the price movement of the underlying stock.
- **Stock + Options Backtesting** : Profit is calculated using the option premium, contract lot size, and trade direction (Buy/Sell).

Although the calculation differs, the final metric always represents the total monetary gain or loss generated by the strategy.

Key Notes

- Net Profit/Loss is an absolute performance metric expressed in monetary terms.
- It is the most direct measure of a strategy's profitability.
- A higher Net Profit does not necessarily indicate a better strategy if it is accompanied by excessive drawdowns or inconsistent returns.

💡 Did You Know?

A strategy with a lower Net Profit can often be superior if it achieves similar returns with significantly lower risk and more consistent performance. Professional traders prioritize risk-adjusted returns, not just absolute profits.

Number of Trades

Definition

Number of Trades represents the total count of completed trades executed by the strategy during the selected backtesting period. Each completed entry and exit pair is considered one trade.

Purpose

This metric indicates how actively the strategy trades and helps evaluate whether the strategy is high-frequency or low-frequency in nature.

Key Notes

- Only completed trades are included in the count.
- A higher number of trades does not necessarily indicate a better strategy.
- More trades provide a larger sample size, making performance metrics generally more reliable.

Win Rate

Definition

Win Rate represents the percentage of completed trades that resulted in a profit during the selected backtesting period. It measures how frequently a trading strategy generates winning trades.

Formula

Win Rate (%) = Winning Trades / Total Trades×100

Purpose

Win Rate helps evaluate the consistency of a trading strategy by showing how often it produces profitable trades. It is useful for comparing different strategies but should always be analysed alongside profitability metrics.

Limitations

- Does not indicate the size of profits or losses.
- A high Win Rate may still result in an unprofitable strategy if losses are larger than gains.
- Should always be evaluated together with Risk-Reward Ratio, Profit Factor, and Expectancy.

Key Notes

- Expressed as a percentage (%).
- Measures the probability of a trade being profitable, not the overall profitability of the strategy.
- Professional traders focus on balancing Win Rate with Risk-Reward Ratio rather than maximizing Win Rate alone.

💡 Did You Know?

A strategy with a 40% Win Rate and a 1:3 Risk-Reward Ratio can often outperform a strategy with an 80% Win Rate but a 1:0.5 Risk-Reward Ratio. Long-term profitability depends on the combination of Win Rate and average profit per trade, not just how often trades win.

Expectancy

Definition

Expectancy measures the average amount a trading strategy is expected to gain or lose per trade over the long run. It combines both the Win Rate and the average profit/loss per trade, making it one of the most reliable indicators of a strategy's long-term profitability.

A positive Expectancy indicates that the strategy is expected to generate profits over a large number of trades, while a negative Expectancy suggests that the strategy is likely to lose money over time.

Formula

Expectancy = (Win Rate×Average Winning Trade) - (Loss Rate×Average Losing Trade)

where,
Loss Rate = 1-Win Rate

Purpose

Expectancy evaluates the long-term profitability of a trading strategy by considering both the probability of winning and the average size of winning and losing trades. It helps determine whether a strategy has a statistical edge over repeated executions.

Limitations

- Based on historical backtesting and does not guarantee future performance.
- Can be influenced by a small number of unusually large winning or losing trades.
- Should be analysed together with Profit Factor, Maximum Drawdown, Win Rate, and Risk-Reward Ratio.

Key Notes

- Expressed as the average profit or loss per trade.
- Considers both the frequency and magnitude of winning and losing trades.
- A strategy with a lower Win Rate can still achieve a high Expectancy if its average winning trades are significantly larger than its average losses.
- Expectancy is one of the most important metrics for evaluating a strategy's long-term statistical advantage.

💡 Did You Know?

Two strategies can have the same Win Rate but completely different Expectancy values. This happens because Expectancy considers not only how often a strategy wins, but also how much it wins or loses on each trade. For this reason, professional traders often rely on Expectancy rather than Win Rate alone when comparing trading strategies.

Profit Factor

Definition

Profit Factor measures the profitability of a trading strategy by comparing the total profit generated from winning trades to the total loss incurred from losing trades. It indicates how many units of profit are earned for every unit of loss.

A higher Profit Factor generally reflects a more profitable and efficient trading strategy.

Formula

Profit Factor = Gross Profit / Gross Loss

where,
Gross Profit = Sum of profits from all winning trades.
Gross Loss = Sum of losses from all losing trades (absolute value).

Purpose

Profit Factor evaluates the overall efficiency of a trading strategy by measuring how effectively it converts losses into profits. It helps traders compare different strategies regardless of the total number of trades executed.

Interpretation

Profit Factor vs Interpretation

Less than 1.0 : Unprofitable strategy (losses exceed profits).
1.0 : Break-even strategy.
1.2 – 1.5 : Acceptable performance.
1.5 – 2.0 : Good and reliable strategy.
Above 2.0 : Excellent profitability and trading efficiency.

A Profit Factor greater than 1 indicates that the strategy generates more profit than loss over the backtesting period.

Limitations

- Does not consider the sequence or consistency of trades.
- Can appear artificially high if generated from a very small number of trades.
- Does not account for drawdowns or volatility.
- Should be analysed together with Expectancy, Maximum Drawdown, Win Rate, and Sharpe Ratio.

Key Notes

- Profit Factor is a ratio, not a percentage.
- A value below 1 indicates that the strategy is losing money overall.
- Higher values generally indicate better profitability, but extremely high values (e.g., above 4) may indicate overfitting or insufficient backtesting data.
- A reliable strategy should maintain a strong Profit Factor across different market conditions and over a sufficiently large number of trades.

💡 Did You Know?

A strategy with a Profit Factor of 2.0 means it generates ₹2 of gross profit for every ₹1 of gross loss. However, this does not guarantee a high Win Rate—a strategy can have a relatively low Win Rate and still achieve an excellent Profit Factor if its winning trades are significantly larger than its losing trades.

1. Expectancy answers : **"How much can I expect to make on average per trade?"**
2. Profit Factor answers : **"How efficiently does my strategy convert losses into profits?"**

Risk / Reward

Definition

Risk/Reward Ratio measures the relationship between the potential loss (risk) and the potential profit (reward) of a trade. It indicates how much profit is expected for every unit of risk taken.

A higher Risk/Reward Ratio generally allows a strategy to remain profitable even with a lower Win Rate.

Formula

Risk/Reward Ratio = Average Winning Trade / Average Losing Trade

Purpose

Risk/Reward Ratio helps evaluate whether the potential reward of a strategy justifies the amount of risk being taken. It is a key metric for assessing trade quality and long-term sustainability.

Limitations

- Does not consider the probability of winning a trade.
- A high Risk/Reward Ratio alone does not guarantee profitability if the Win Rate is too low.
- Should always be analysed together with Win Rate, Expectancy, and Profit Factor.

Key Notes

- Expressed as a ratio, not a percentage.
- Higher values generally indicate better reward relative to risk.
- The ideal Risk/Reward Ratio depends on the trading strategy and market conditions.
- A balanced combination of Win Rate and Risk/Reward Ratio is essential for long-term profitability.

💡 Did You Know?

A strategy with a 40% Win Rate and a 3:1 Risk/Reward Ratio can be more profitable than a strategy with an 80% Win Rate and a 0.5:1 Risk/Reward Ratio. Successful trading depends on the balance between how often you win and how much you win when you do.

Average Winning Trade

Definition

Average Winning Trade represents the average profit earned from all profitable trades executed during the selected backtesting period. It indicates how much profit, on average, each winning trade generates.

Formula

Average Winning Trade =Gross Profit / Number of Winning Trades

Purpose

This metric helps evaluate the average profitability of successful trades and is used to assess the reward potential of a trading strategy.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Includes only profitable trades.
- Used in the calculation of Expectancy and Risk-Reward Ratio.
- A higher value is generally desirable, provided losses remain controlled.

💡 Did You Know?

A strategy with a low Win Rate can still be highly profitable if its Average Winning Trade is significantly larger than its Average Losing Trade. This is why professional traders focus on both the size and frequency of winning trades, rather than Win Rate alone.

Average Losing Trade

Definition

Average Losing Trade represents the average loss incurred from all losing trades executed during the selected backtesting period. It indicates how much capital, on average, is lost on each unsuccessful trade.

Formula

Average Losing Trade =Gross Loss / Number of Losing Trades

Purpose

This metric helps evaluate the average size of losing trades and is used to assess the risk associated with a trading strategy.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Includes only losing trades.
- Used in the calculation of Expectancy and Risk-Reward Ratio.
- Lower values are generally desirable, provided they do not significantly reduce the strategy's profit potential.

💡 Did You Know?

Two strategies can have the same Net Profit, but the one with a lower Average Losing Trade is generally considered more robust because it manages risk more effectively and reduces the impact of losing streaks.

CAGR

Definition

Compound Annual Growth Rate (CAGR) represents the average annual growth rate of the trading capital over the backtesting period, assuming that profits are reinvested (compounded) each year.

It provides a standardized measure of investment growth, making it easier to compare strategies with different backtesting durations.

Formula

CAGR (%) = $((\text{Final Capital} / \text{Initial Capital})^{(1/n)}) - 1$

where,

Final Capital = Capital at the end of the backtest.
Initial Capital = Starting capital.
n = Backtesting period (in years).

Purpose

CAGR measures the annualized growth of a trading strategy, allowing traders to compare the long-term performance of different strategies on a consistent basis.

Limitations

- Assumes a smooth annual growth rate, whereas actual trading returns often fluctuate.
- Does not account for volatility or drawdowns.
- Should be analysed together with Maximum Drawdown, Sharpe Ratio, and Calmar Ratio.

Key Notes

- Expressed as an annual percentage (%).
- Enables fair comparison between strategies tested over different time periods.
- Higher CAGR is generally desirable but should always be evaluated alongside risk-adjusted performance metrics.

💡 Did You Know?

Two strategies may achieve the same overall return, but the one that reaches it in a shorter time will have a higher CAGR. This makes CAGR one of the most useful metrics for comparing long-term strategy performance across different backtesting periods.

Sharpe Ratio

Definition

Sharpe Ratio measures the risk-adjusted return of a trading strategy by comparing its average excess return to the volatility of those returns. It indicates how much return the strategy generates for each unit of risk taken.

A higher Sharpe Ratio suggests that the strategy delivers more consistent returns relative to its level of risk.

Formula

Sharpe Ratio = $(R_p - R_f) / \sigma_p$

where,

- R_p = Average return of the strategy.
- R_f = Risk-free rate of return.
- σ_p = Standard deviation (volatility) of strategy returns.

Purpose

Sharpe Ratio evaluates the quality of a strategy's returns by considering both profitability and volatility. It helps traders determine whether higher returns are achieved efficiently or simply by taking excessive risk.

Limitations

- Treats both positive and negative return fluctuations as risk.
- Assumes returns follow a normal distribution.
- Can be influenced by extreme market movements.
- Should be evaluated together with Sortino Ratio, Maximum Drawdown, and Calmar Ratio.

Key Notes

- Measures return per unit of total risk.
- Higher values indicate better risk-adjusted performance.
- Useful for comparing multiple trading strategies with different return and volatility characteristics.
- One of the most widely used performance metrics in quantitative finance and portfolio management.

💡 Did You Know?

Two strategies may generate the same annual return, but the one with lower volatility will have a higher Sharpe Ratio. This is why professional traders evaluate not only how much a strategy earns, but also how consistently it earns those returns.

Sortino Ratio

Definition

Sortino Ratio measures the risk-adjusted return of a trading strategy by comparing its average excess return to downside volatility. Unlike the Sharpe Ratio, it considers only negative return fluctuations as risk, making it a more accurate measure for evaluating trading strategies.
A higher Sortino Ratio indicates that the strategy generates higher returns while minimizing downside risk.

Formula

Sharpe Ratio = (Rp-Rf) / Rf

where,

- Rp = Average return of the strategy.
- Rf = Risk-free rate of return.
- σp = Downside deviation (standard deviation of negative returns)

Purpose

Sortino Ratio evaluates how efficiently a trading strategy generates returns while considering only harmful volatility. It helps traders assess performance without penalizing positive return fluctuations.

Limitations

- Considers only downside volatility and ignores overall return variability.
- Like all historical metrics, it does not guarantee future performance.
- Should be analysed together with Sharpe Ratio, Maximum Drawdown, and Calmar Ratio.

Key Notes

- Measures return per unit of downside risk.
- Penalizes only negative return fluctuations.
- Particularly useful for evaluating trading strategies where upside volatility is considered desirable.
- Generally preferred over the Sharpe Ratio when focusing on downside risk.

💡 Did You Know?

The Sharpe Ratio treats both positive and negative volatility as risk, whereas the Sortino Ratio penalizes only negative volatility. This makes the Sortino Ratio especially useful for trading strategies that experience large positive price movements but maintain controlled downside risk.

Calmar Ratio

Definition

Calmar Ratio measures the risk-adjusted performance of a trading strategy by comparing its annualized return (CAGR) to its Maximum Drawdown. It indicates how much return the strategy generates for each unit of downside risk.

A higher Calmar Ratio suggests that the strategy delivers stronger returns while maintaining lower drawdowns.

Formula

Calmar Ratio = CAGR / Maximum Drawdown

where,
CAGR = Compound Annual Growth Rate.
Maximum Drawdown = Largest peak-to-trough decline in portfolio value during the backtesting period.

Purpose

Calmar Ratio evaluates how efficiently a trading strategy generates returns relative to its worst historical drawdown. It is particularly useful for comparing strategies with different levels of downside risk.

Limitations

- Relies solely on Maximum Drawdown, ignoring overall return volatility.
- Based on historical performance and does not guarantee future results.
- Should be analysed together with Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.

Key Notes

- Measures return per unit of maximum drawdown.
- Particularly useful for evaluating trend-following and systematic trading strategies.
- Higher values generally indicate a more efficient balance between return and downside risk.

💡 Did You Know?

While the Sharpe Ratio measures returns relative to volatility and the Sortino Ratio considers downside volatility, the Calmar Ratio focuses exclusively on the strategy's largest historical drawdown. This makes it especially valuable for traders who prioritize capital preservation and drawdown control.

Maximum Drawdown

Definition

Maximum Drawdown (MDD) represents the largest percentage decline in portfolio value from a peak to a subsequent trough during the selected backtesting period, before a new peak is reached.

It measures the maximum capital loss an investor would have experienced if they had started investing at the highest point and exited at the lowest point.

Formula

Maximum Drawdown (%) = ((Peak Portfolio Value - Trough Portfolio Value) / Peak Portfolio Value)×100

where,

Peak Portfolio Value = Highest portfolio value before the decline.
Trough Portfolio Value = Lowest portfolio value before a new peak is achieved.

Purpose

Maximum Drawdown measures the largest historical loss experienced by a trading strategy and helps evaluate its downside risk. It is one of the most important metrics for assessing capital preservation and the overall risk associated with a strategy.

Limitations

- Based solely on historical performance and may not reflect future risk.
- Measures only the largest loss, ignoring the frequency and duration of smaller drawdowns.
- Should be analysed together with Net Profit, CAGR, Sharpe Ratio, Sortino Ratio, and Calmar Ratio.

Key Notes

- Expressed as a percentage (%).
- Measures the worst historical decline in portfolio value.
- Lower values are generally preferred as they indicate better capital preservation.
- One of the most important metrics for evaluating the risk of a trading strategy.

💡 Did You Know?

Two strategies may generate the same Net Profit, but the one with the lower Maximum Drawdown is generally considered superior because it achieves similar returns while exposing the trader to significantly less downside risk. This is why professional traders often prioritize drawdown control over maximizing returns alone.

Maximum Winning Trade

Definition

Maximum Winning Trade represents the highest profit earned from a single completed trade during the selected backtesting period. It highlights the strategy's most profitable individual trade.

Formula

Maximum Winning Trade = max(Profit of All Winning Trades)

Purpose

This metric helps identify the largest profit generated from a single trade and provides insight into the strategy's maximum earning potential.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Considers only profitable trades.
- Useful for identifying the strategy's best-performing trade.
- Should not be used in isolation to evaluate overall strategy performance.

💡 Did You Know?

A strategy with an exceptionally high Maximum Winning Trade but a low Average Winning Trade may be relying on a few outlier trades rather than generating consistent profits across multiple trades.

Maximum Losing Trade

Definition

Maximum Losing Trade represents the largest loss incurred from a single completed trade during the selected backtesting period. It highlights the strategy's worst-performing individual trade.

Formula

Maximum Losing Trade = max(Loss of All Losing Trades)

Purpose

This metric helps identify the largest loss generated from a single trade and provides insight into the worst-case trade experienced by the strategy.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Considers only losing trades.
- Useful for identifying the strategy's worst-performing trade.
- Should not be used in isolation to evaluate overall strategy performance.

💡 Did You Know?

A strategy with a small Average Losing Trade but a very large Maximum Losing Trade may indicate poor risk management or the absence of an effective stop-loss, where a single trade can significantly impact overall performance.

Average Holding Time

Definition

Average Holding Time represents the average duration for which a trade remains open before it is closed during the selected backtesting period. It indicates how long, on average, the strategy holds positions.

Formula

Average Holding Time = ∑ Holding Time of All Trades / Total Number of Trades

Purpose

This metric helps identify the typical trade duration and provides insight into the trading style, whether it is scalping, intraday, swing trading, or long-term investing.

Key Notes

- Expressed in seconds, minutes, hours, or days, depending on the strategy.
- Calculated using all completed trades.
- Helps classify the trading style and evaluate execution efficiency.
- Should be interpreted in the context of the strategy's objectives rather than in isolation.

💡 Did You Know?

A short Average Holding Time does not necessarily indicate a better strategy. Many successful trend-following strategies deliberately hold trades for longer periods to capture larger price movements, while scalping strategies aim to profit from small moves over much shorter durations.

Winning Trades

Definition

Winning Trades represents the total number of completed trades that resulted in a profit during the selected backtesting period.

Formula

Winning Trades=Count of Profitable Trades

Purpose

This metric indicates how many trades were successfully closed in profit and is used to calculate the Win Rate and evaluate the consistency of a trading strategy.

Key Notes

- Includes only completed profitable trades.
- Expressed as a count, not a percentage.
- Used in the calculation of Win Rate.
- A higher number of winning trades does not necessarily indicate a profitable strategy if the losing trades are significantly larger.

💡 Did You Know?

A strategy can have fewer Winning Trades than Losing Trades and still be highly profitable if the average profit from each winning trade is significantly greater than the average loss from each losing trade.

Losing Trades

Definition

Losing Trades represents the total number of completed trades that resulted in a loss during the selected backtesting period.

Formula

Losing Trades=Count of Losing Trades

Purpose

This metric indicates how many trades were closed at a loss and is used to calculate the Win Rate and evaluate the overall performance of a trading strategy.

Key Notes

- Includes only completed losing trades.
- Expressed as a count, not a percentage.
- Used in the calculation of Win Rate.
- A higher number of losing trades does not necessarily indicate an unprofitable strategy if the average winning trade is significantly larger than the average losing trade.

💡 Did You Know?

Many successful trading strategies have more Losing Trades than Winning Trades. As long as the average winning trade is sufficiently larger than the average losing trade, the strategy can remain consistently profitable.

Gross Profit

Definition

Gross Profit represents the total profit earned from all winning trades during the selected backtesting period, before considering any losses.

Formula

Gross Profit = ∑ Profit from All Winning Trades

Purpose

This metric measures the total amount of profit generated by successful trades and is used to evaluate the earning potential of a trading strategy.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Includes only profitable trades.
- Does not account for losses or transaction costs.
- Used in the calculation of Profit Factor and Net Profit/Loss.

💡 Did You Know?

A strategy can have a high Gross Profit and still lose money overall if its Gross Loss is even larger. This is why Gross Profit should never be evaluated in isolation.

Gross Loss

Definition

Gross Loss represents the total loss incurred from all losing trades during the selected backtesting period, before considering any profits.

Formula

Gross Loss = ∑ Loss from All Losing Trades

Purpose

This metric measures the total amount of loss generated by unsuccessful trades and helps evaluate the downside risk of a trading strategy.

Key Notes

- Expressed in monetary terms (₹, $, €, etc.).
- Includes only losing trades.
- Does not account for profits or transaction costs.
- Used in the calculation of Profit Factor and Net Profit/Loss.

💡 Did You Know?

A strategy can have a high Gross Loss and still be highly profitable if its Gross Profit is significantly larger. The balance between these two metrics determines the strategy's overall profitability, which is reflected by the Profit Factor.

Monthly Statistics

Definition

Monthly Statistics provides a month-wise summary of the trading strategy's performance over the selected backtesting period. It helps traders analyze consistency, identify profitable and unprofitable months, and evaluate how the strategy performs under different market conditions.

Purpose

This table enables traders to:

- Evaluate month-by-month strategy performance.
- Identify consistently profitable or loss-making periods.
- Detect seasonal trends and changing market behavior.
- Assess the stability and consistency of the trading strategy over time.

Key Notes

- Statistics are calculated separately for each calendar month.
- Enables easy comparison of performance across different months and years.
- Useful for identifying seasonal trends and evaluating long-term consistency.
- Should be analysed alongside overall performance metrics such as Net Profit, Maximum Drawdown, and CAGR.

💡 Did You Know?

A strategy with strong overall profitability may still have several losing months. Analyzing Monthly Statistics helps determine whether the strategy is consistently profitable or relies heavily on a few exceptional months.

Day Statistics

Definition

Day-wise Statistics provides a summary of the trading strategy's performance based on the day of the week (Monday to Friday). It helps identify whether the strategy performs better or worse on specific trading days.

Purpose

This table enables traders to:

- Analyze performance across different trading days.
- Identify the strongest and weakest trading days.
- Detect recurring patterns in weekly market behavior.
- Optimize trading by focusing on historically stronger days.

Key Notes

- Statistics are grouped by day of the week.
- Useful for identifying recurring weekly performance patterns.
- Helps evaluate the consistency of the strategy across different trading days.
- Should be analysed alongside Monthly Statistics and overall performance metrics.

💡 Did You Know?

Many trading strategies perform better on specific days of the week due to recurring market behavior, such as higher volatility on expiry days or increased institutional activity at the beginning of the trading week.

Hour-wise Statistics

Definition

Hour-wise Statistics provides a summary of the trading strategy's performance based on the hour during which trades were executed. It helps identify the most and least profitable trading hours within the trading session.

Purpose

This table enables traders to:

- Analyze performance during different trading hours.
- Identify periods with the highest profitability.
- Detect time-based market behavior and volatility.
- Optimize trading by focusing on historically stronger trading hours.

Key Notes

- Statistics are grouped by the trade execution hour.
- Useful for identifying the most favorable trading periods.
- Helps optimize trade timing and avoid historically weaker periods.
- Should be analysed alongside Day-wise Statistics and overall performance metrics.

💡 Did You Know?

Market behavior changes throughout the trading day. The opening session often experiences high volatility, midday trading is typically calmer, and volatility may increase again toward the market close. Analyzing Hour-wise Statistics helps identify the periods where a strategy performs most consistently.

Equity Curve

Definition

The Equity Curve is a graphical representation of the trading account's cumulative value over the selected backtesting period. It illustrates how the portfolio value changes after each completed trade, providing a visual summary of the strategy's overall performance.

Purpose

The Equity Curve enables traders to:

- Visualize the growth of trading capital over time.
- Evaluate the consistency and stability of strategy performance.
- Identify periods of rapid growth, stagnation, or drawdowns.
- Assess the long-term profitability and reliability of the trading strategy.

Interpretation

Equity Curve Behavior vs Interpretation

Smooth Upward Trend : Consistent and stable profitability
Steep Upward Trend : Rapid capital growth with strong performance
Sideways Movement : Little or no portfolio growth
Frequent Large Declines : High volatility and increased trading risk
Downward Trend : Strategy is consistently losing capital

Key Notes

- Displays the cumulative portfolio value after each completed trade.
- Provides a visual representation of both returns and drawdowns.
- A smoother equity curve generally indicates a more stable and reliable strategy.
- Should be analysed alongside Net Profit, Maximum Drawdown, Sharpe Ratio, and Calmar Ratio.

💡 Did You Know?

Two strategies can generate the same Net Profit, but the one with the smoother Equity Curve is generally considered superior because it achieves similar returns with lower volatility and smaller drawdowns. Professional traders often prefer strategies with consistent equity growth over those with erratic performance, even if the final profits are similar.

Drawdown Curve

Definition

The Drawdown Curve is a graphical representation of the percentage decline in portfolio value from its previous peak over the selected backtesting period. It illustrates the magnitude and duration of losses experienced before the portfolio recovers to a new high.

Purpose

The Drawdown Curve enables traders to:

- Visualize periods of capital decline.
- Identify the severity and duration of drawdowns.
- Evaluate the risk and stability of the trading strategy.
- Assess how quickly the strategy recovers after losses.

Interpretation

Drawdown Curve Behavior vs Interpretation

Small, Short Drawdowns : Indicates good capital preservation and faster recovery
Deep Drawdowns : Indicates higher downside risk and larger capital losses
Long Recovery Periods : Strategy takes longer to recover from losses
Frequent Drawdowns : Strategy experiences repeated periods of capital decline

Key Notes

- Displayed as a percentage (%) below the portfolio's peak value.
- A value of 0% indicates the portfolio is at a new all-time high.
- Larger negative values indicate deeper drawdowns.
- Complements the Equity Curve by highlighting periods of capital decline.
- Should be analysed alongside Maximum Drawdown, Calmar Ratio, and Net Profit.

💡 Did You Know?

Two strategies may have the same Maximum Drawdown, but their Drawdown Curves can look very different. One strategy may recover within a few days, while another may take several months to regain its previous peak. This makes the Drawdown Curve an excellent tool for assessing not only how much a strategy loses, but also how long it takes to recover.

Trade Histogram

Definition

The Trade Histogram is a chart that displays the distribution of individual trade results by grouping profits and losses into value ranges (bins). It shows how frequently trades fall within each profit or loss range.

Purpose

The Trade Histogram enables traders to:

- Visualize the distribution of trade outcomes.
- Identify whether profits and losses are concentrated in specific ranges.
- Detect outlier trades with unusually large gains or losses.
- Assess the consistency and variability of trade performance.

Interpretation

Histogram Pattern vs Interpretation

Cluster around small profits : Strategy generates frequent modest gains
Long right tail : Occasional large winning trades significantly contribute to profits
Long left tail : Occasional large losing trades increase downside risk
Symmetric distribution : Profits and losses are more evenly distributed

Key Notes

- Each bar represents the number of trades within a specific profit/loss range.
- The horizontal axis shows trade profit or loss.
- The vertical axis shows the frequency of trades.
- Useful for identifying skewness, outliers, and the overall shape of trade outcomes.
- Should be analysed alongside Average Winning Trade, Average Losing Trade, Expectancy, and Profit Factor.

💡 Did You Know?

A strategy can be profitable even if most trades are small losses, provided the histogram shows a long right tail containing a few large winning trades that outweigh the cumulative losses. The shape of the histogram often reveals the true profit distribution more clearly than the Win Rate alone.

Monthly Returns Heatmap

Definition

A strategy can be profitable even if most trades are small losses, provided the histogram shows a long right tail containing a few large winning trades that outweigh the cumulative losses. The shape of the histogram often reveals the true profit distribution more clearly than the Win Rate alone.

Purpose

The Monthly Returns Heatmap enables traders to:

- Analyze month-wise performance over multiple years.
- Identify consistently profitable or loss-making months.
- Detect seasonal trends and recurring market patterns.
- Evaluate the consistency of the trading strategy over time.

Key Notes

- Each cell represents the percentage return for a specific month.
- Colors indicate the magnitude of monthly gains or losses.
- Enables quick comparison of performance across different months and years.
- Useful for identifying seasonal trends and evaluating long-term consistency.
- Should be analysed alongside Monthly Statistics, Equity Curve, and Drawdown Curve.

💡 Did You Know?

A strategy may have an excellent overall CAGR, yet the Monthly Returns Heatmap can reveal recurring weak months or seasonal patterns. Identifying these trends can help traders better understand strategy behavior and optimize trading during historically stronger market periods.

Rolling Win Rate(20 Trade Window)

Definition

The Rolling Win Rate (20-Trade Window) is a moving performance metric that calculates the Win Rate over the most recent 20 completed trades. As each new trade is completed, the oldest trade in the window is removed and the newest trade is included, providing a continuously updated view of the strategy's recent performance.

Purpose

The Rolling Win Rate enables traders to:

- Monitor short-term changes in strategy performance.
- Identify periods of improving or deteriorating consistency.
- Detect shifts in market conditions affecting the strategy.
- Evaluate whether the strategy is maintaining its historical performance.

Key Notes

- Calculated using a rolling window of the most recent 20 completed trades.
- Updates after every completed trade.
- Provides a dynamic view of recent strategy performance instead of the overall historical Win Rate.
- Useful for detecting performance changes that may be hidden by the overall Win Rate.
- Should be analysed alongside Win Rate, Expectancy, and Profit Factor.

💡 Did You Know?

A strategy may have an overall Win Rate of 60%, but its Rolling Win Rate can temporarily fall below 40% or rise above 80%. This makes the Rolling Win Rate a valuable tool for identifying periods where the strategy is outperforming or underperforming its long-term average.

Holding Time vs P&L

Definition

The Holding Time vs P&L chart visualizes the relationship between a trade's holding duration and its corresponding profit or loss (P&L). Each point on the chart represents a completed trade, helping traders analyze whether trade duration has an impact on profitability.

Purpose

The Holding Time vs P&L chart enables traders to:

- Analyze the relationship between trade duration and profitability.
- Identify the optimal holding period for the strategy.
- Detect whether trades held longer tend to generate larger profits or losses.
- Evaluate the efficiency of trade exits.

Key Notes

- Each point represents a single completed trade.
- The X-axis represents the trade holding time.
- The Y-axis represents the corresponding Profit/Loss (P&L).
- Helps identify the strategy's optimal holding duration.
- Should be analysed alongside Average Holding Time, Maximum Winning Trade, and Maximum Losing Trade.

💡 Did You Know?

Many profitable traders follow the principle of "Cut losses quickly and let profits run." If this principle is reflected in the chart, losing trades will typically cluster around short holding times, while profitable trades will be spread across longer holding durations, indicating disciplined risk management and effective trade exits.

Trade Log

Definition

The Trade Log is a detailed record of every trade executed during the selected backtesting period. Each row represents a completed trade and includes all relevant information, such as entry and exit details, trade direction, holding time, profit/loss, and other trade-specific metrics.

Purpose

The Trade Log enables traders to:

- Review every executed trade individually.
- Verify the accuracy of backtesting results.
- Analyze entry and exit decisions.
- Identify recurring trading patterns and mistakes.
- Export trade data for further analysis.

Key Notes

- Each row represents one completed trade.
- Includes detailed trade information such as entry, exit, holding time, and profit/loss.
- Serves as the primary source for calculating all backtesting performance metrics.
- Can be filtered, sorted, and exported for further analysis.
- Useful for validating strategy behavior and identifying opportunities for improvement.

💡 Did You Know?

Professional traders spend as much time reviewing their Trade Log as they do placing trades. Analyzing individual trades often reveals recurring strengths and weaknesses that may not be visible through summary metrics alone.

THANK YOU!

This project was built to bring together market data, technical indicators, trading strategies, backtesting, performance analysis, and options-based strategies into one practical system.

More than just a project, Project Quant represents an ongoing effort to understand how data, logic, and systematic decision-making can be applied to financial markets.

Thank you for taking the time to explore Project Quant.